

# **LAPORAN PRAKTIKUM PERTEMUAN 3 PEMROGRAMAN MOBILE**

## **Form Input dan State Management**



### **Disusun oleh:**

Nama : Fadhil Afiq Badruzzaman

NIM : 20240801119

Prodi : Teknik Informatika

### **Dosen Pengampu:**

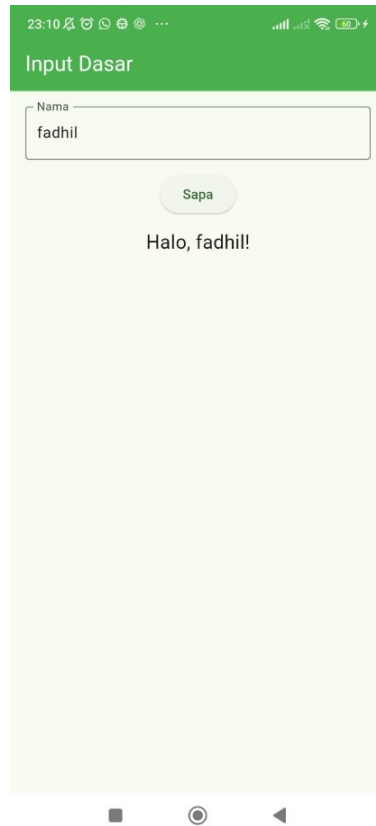
JEFRY SUNUPURWA ASRI , S.Kom., M.Kom.

**Universitas Esa Unggul Tangerang**

**2026**

## MODUL 3

## Bagian A Input Dasar



## Code

```
1 import 'package:flutter/material.dart';
2
3 void main() => runApp(const MyApp());
4
5 class MyApp extends StatelessWidget {
6   const MyApp({super.key});
7
8   @override
9   Widget build(BuildContext context) {
10    return MaterialApp(
11      debugShowCheckedModeBanner: false,
12      title: 'Praktikum 3',
13      theme: ThemeData(
14        colorSchemeSeed: Colors.green,
15        useMaterial3: true,
16      ),
17      home: const InputPage(),
18    );
19  }
20 }
21
22 class InputPage extends StatefulWidget {
23   const InputPage({super.key});
24
25   @override
26   State<InputPage> createState() => _InputPageState();
27 }
28
29 class _InputPageState extends State<InputPage> {
30   final _controller = TextEditingController();
31   String _salam = '';
32
33   @override
34   void dispose() {
35     _controller.dispose();
36     super.dispose();
37   }
38
39   @override
40   Widget build(BuildContext context) {
41     return Scaffold(
42       appBar: AppBar(
43         title: const Text('Input Dasar'),
44         backgroundColor: Colors.green,
45         foregroundColor: Colors.white,
46       ),
47       body: Padding(
48         padding: const EdgeInsets.all(16),
49         child: Column(
50           children: [
51             TextField(
52               controller: _controller,
53               decoration: const InputDecoration(
54                 labelText: 'Nama',
55                 border: OutlineInputBorder(),
56               ),
57             ),
58             const SizedBox(height: 12),
59             ElevatedButton(
60               onPressed: () {
61                 setState(() {
62                   _salam = 'Halo, ${_controller.text}!';
63                 });
64               },
65               child: const Text('Sapa'),
66             ),
67             const SizedBox(height: 12),
68             Text(
69               _salam,
70               style: const TextStyle(fontSize: 20),
71             ),
72           ],
73         ),
74       ),
75     );
76   }
77 }
```

## Penjelasan

|                      |                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| StatelessWidget      | Widget yang tampilannya <b>tidak memiliki state yang berubah</b> selama widget tersebut aktif.                                          |
| StatefulWidget       | Widget yang memiliki <b>state/data yang dapat berubah</b> , sehingga tampilan dapat diperbarui.                                         |
| State<InputPage>     | Menentukan class state yang mengatur data dan perubahan tampilan milik InputPage.                                                       |
| BuildContext context | Menunjukkan posisi sebuah widget di dalam <b>widget tree</b> Flutter dan digunakan untuk mengakses informasi/widget lain di sekitarnya. |
| MaterialApp          | Widget utama aplikasi berbasis Material Design yang mengatur tema, halaman awal, navigasi, dan konfigurasi aplikasi.                    |
| ThemeData(...)       | Mengatur tema global aplikasi seperti warna dan gaya Material Design.                                                                   |
| dispose()            | Method lifecycle yang dijalankan ketika widget sudah tidak digunakan, biasanya untuk membersihkan controller/resource.                  |
| Scaffold             | Menyediakan struktur dasar satu halaman Flutter, seperti AppBar, body, FloatingActionButton, dan lainnya.                               |
| InputDecoration      | Mengatur tampilan dan dekorasi sebuah input seperti label, border, hint, dan icon.                                                      |

### Bagian B Form dengan Validasi

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Form Pendaftaran

Nama lengkap

Fadhil Afiq Badruzzaman

Email

fadhil@gmail.com

Jurusan

Teknik Informatika

☒ Saya menyetujui ketentuan

Daftar

Terdaftar: Fadhil Afiq Badruzzaman (T1)

## Code

```

1 import 'package:flutter/material.dart';
2
3 void main() => runApp(const MyApp());
4
5 class MyApp extends StatelessWidget {
6   const MyApp({super.key});
7
8   @override
9   Widget build(BuildContext context) {
10     return MaterialApp(
11       debugShowCheckedModeBanner: false,
12       title: 'Praktikum 3',
13       theme: ThemeData(
14         colorSchemeSeed: Colors.green,
15         useMaterial3: true,
16       ),
17       home: const FormPage(),
18     );
19   }
20 }
21
22 class FormPage extends StatefulWidget {
23   const FormPage({super.key});
24
25   @override
26   State<FormPage> createState() => FormPageState();
27 }
28
29 class FormPageState extends State<FormPage> {
30   final _formKey = GlobalKey<FormState>();
31
32   final _nama = TextEditingController();
33   final _email = TextEditingController();
34
35   String? _jurusan;
36   bool _setuju = false;

```

```

79         if (value == null || value.trim().isEmpty()) {
80             return 'Nama wajib diisi';
81         }
82     },
83     return null;
84 },
85 ),
86
87 const SizedBox(height: 12),
88
89 TextFormField(
90     controller: _email,
91     keyboardType: TextInputType.emailAddress,
92     decoration: const InputDecoration(
93         labelText: 'Email',
94         border: OutlineInputBorder(),
95     ),
96     validator: (value) {
97         if (value == null || !value.contains('@')) {
98             return 'Email tidak valid';
99         }
100     },
101     return null;
102 },
103 ),
104
105 const SizedBox(height: 12),
106
107 DropdownButtonFormField<String>{
108     decoration: const InputDecoration(
109         labelText: 'Jurusan',
110         border: OutlineInputBorder(),
111     ),
112     items: const [
113         DropdownMenuItem(
114             value: 'TI',
115             child: Text('Teknik Informatika'),
116         ),
117         DropdownMenuItem(
118             value: 'SI',
119             child: Text('Sistem Informasi'),

```

```

37
38 @override
39 void dispose() {
40   nama.dispose();
41   _email.dispose();
42   super.dispose();
43 }
44
45 void _kirim() {
46   if (_formKey.currentState!.validate()) {
47     final jurusan = _jurusan ?? '-';
48
49     ScaffoldMessenger.of(context).showSnackBar(
50       SnackBar(
51         content: Text(
52           'Terdaftar: ${_nama.text} ($jurusan)',
53         ),
54       ),
55     );
56   }
57 }
58
59 @override
60 Widget build(BuildContext context) {
61   return Scaffold(
62     appBar: AppBar(
63       title: const Text('Form Pendaftaran'),
64       backgroundColor: Colors.green,
65       foregroundColor: Colors.white,
66     ),
67     body: Form(
68       key: _formKey,
69       child: ListView(
70         padding: const EdgeInsets.all(16),
71         children: [
72           TextFormField(
73             controller: _nama,
74             decoration: const InputDecoration(
75               labelText: 'Nama lengkap',
76               border: OutlineInputBorder(),
77             ),
78             validator: (value) {

```

```

120       ),
121       DropdownMenuItem(
122         value: 'IE',
123         child: Text('Teknik Elektro'),
124       ),
125     ],
126     onChanged: (value) {
127       setState(() {
128         _jurusan = value;
129       });
130     },
131     validator: (value) {
132       if (value == null) {
133         return 'Pilih jurusan';
134       }
135
136       return null;
137     },
138   ),
139
140   CheckboxListTile(
141     title: const Text('Saya menyetujui ketentuan'),
142     value: _setuju,
143     activeColor: Colors.green,
144     controlAffinity: ListTileControlAffinity.leading,
145     onChanged: (value) {
146       setState(() {
147         _setuju = value ?? false;
148       });
149     },
150   ),
151
152   ElevatedButton(
153     onPressed: _setuju ? _kirim : null,
154     child: const Text('Daftar'),
155   ),
156 ],
157 ),
158 );
159 }
160 }
161 }

```

## Penjelasan

|                                                       |                                                                                                                 |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <code>_formKey.currentState!.validate()</code>        | Menjalankan seluruh fungsi validator pada field yang ada di dalam Form. Jika semua valid, hasilnya true.        |
| <code>validator: (value) { ... }</code>               | Fungsi untuk mengecek apakah nilai input pengguna memenuhi aturan tertentu.                                     |
| <code>value.trim()</code>                             | Menghapus spasi di awal dan akhir input sebelum dilakukan pengecekan.                                           |
| <code>keyboardType: TextInputType.emailAddress</code> | Mengatur jenis keyboard perangkat agar lebih sesuai untuk input email, biasanya menampilkan karakter seperti @. |
| <code>onChanged: (value) { ... }</code>               | Event yang dijalankan ketika pengguna memilih atau mengubah nilai suatu input.                                  |
| <code>ScaffoldMessenger.of(context)</code>            | Digunakan untuk menampilkan pesan sementara seperti SnackBar pada halaman Flutter.                              |
| <code>_setuju = value ?? false</code>                 | Jika value bernilai null, maka otomatis menggunakan nilai false. Operator ?? disebut null-coalescing operator   |

## Perlu flutter add provider buat bagian D

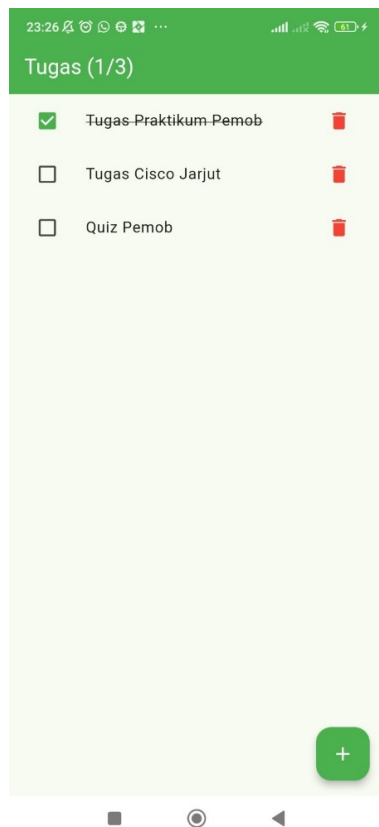
```
fadhilbzm@FIQRI:~/perkuliahan/pemrograman_mobile/praktikum_1$ flutter pub add provider
Resolving dependencies...
Downloading packages... (2.7s)
  async 2.11.0 (2.13.1 available)
  boolean_selector 2.1.1 (2.1.2 available)
  characters 1.3.0 (1.4.1 available)
  clock 1.1.1 (1.1.3 available)
  collection 1.18.0 (1.19.1 available)
  cupertino_icons 1.0.8 (2.0.0 available)
  fake_async 1.3.1 (1.3.3 available)
  flutter_lints 4.0.0 (6.0.0 available)
  leak_tracker 10.0.5 (11.0.2 available)
  leak_tracker_flutter_testing 3.0.5 (3.0.10 available)
  leak_tracker_testing 3.0.1 (3.0.2 available)
  lints 4.0.0 (6.1.0 available)
  matcher 0.12.16+1 (0.12.20 available)
  material_color_utilities 0.11.1 (0.13.1 available)
  meta 1.15.0 (1.19.0 available)
+ nested 1.0.0
  path 1.9.0 (1.9.1 available)
+ provider 6.1.5+1
  source_span 1.10.0 (1.10.2 available)
  stack_trace 1.11.1 (1.12.2 available)
  stream_channel 2.1.2 (2.1.4 available)
  string_scanner 1.2.0 (1.4.1 available)
  term_glyph 1.2.1 (1.2.2 available)
  test_api 0.7.2 (0.7.14 available)
  vector_math 2.1.4 (2.4.3 available)
  vm_service 14.2.5 (15.3.0 available)
Changed 2 dependencies!
24 packages have newer versions incompatible with dependency constraints.
Try 'flutter pub outdated' for more information.
fadhilbzm@FIQRI:~/perkuliahan/pemrograman_mobile/praktikum_1$ |
```

## Penjelasan

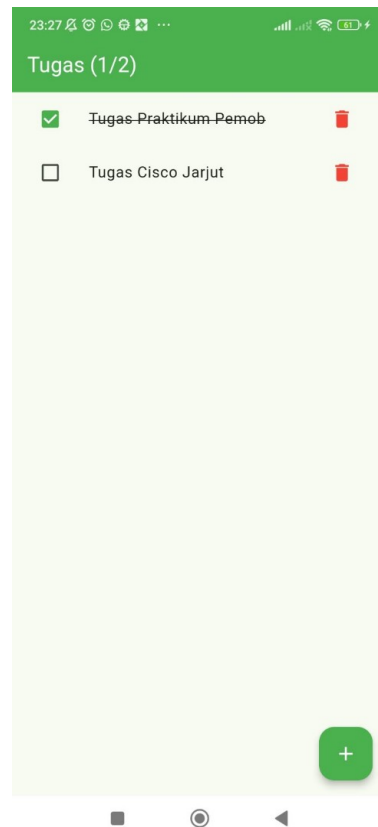
flutter pub add provider itu buat **memasang package Provider ke project Flutter**.

Di **Bagian D**, Provider dipakai karena state/data TugasModel mau dipakai oleh **lebih dari satu halaman**, yaitu TugasPage dan TambahPage. Di modul juga memang diminta memasang package ini sebelum memakai ChangeNotifierProvider, context.watch, dan context.read.

## Bagian D Daftar Tugas dengan Provider



Daftar tugas yang selesai 1 akan di coret



Tugas yang di hapus

## Code

```
1 import 'package:flutter/material.dart';
2 import 'package:provider/provider.dart';
3
4 class Tugas {
5   String judul;
6   bool selesai;
7
8   Tugas(this.judul, {this.selesai = false});
9 }
10
11 class TugasModel extends ChangeNotifier {
12   final List<Tugas> _items = [];
13
14   List<Tugas> get items => List.unmodifiable(_items);
15
16   int get jumlahSelesai =>
17     _items.where((tugas) => tugas.selesai).length;
18
19   void tambah(String judul) {
20     _items.add(Tugas(judul));
21     notifyListeners();
22   }
23
24   void toggle(int index) {
25     _items[index].selesai = !_items[index].selesai;
26     notifyListeners();
27   }
28
29   void hapus(int index) {
30     _items.removeAt(index);
31     notifyListeners();
32   }
33 }
34
35 void main() {
36   runApp(
37     ChangeNotifierProvider(
38       create: (_) => TugasModel(),
39       child: const MyApp(),
40     ),
41   );
42 }
43
44 class MyApp extends StatelessWidget {
45   const MyApp({super.key});
```

```
96     title: Text(
97       tugas.judul,
98       style: TextStyle(
99         decoration: tugas.selesai
100           ? TextDecoration.lineThrough
101           : null,
102       ),
103     ),
104     trailing: IconButton(
105       icon: const Icon(
106         Icons.delete,
107         color: Colors.red,
108       ),
109       onPressed: () {
110         context.read<TugasModel>().hapus(index);
111       },
112     ),
113   );
114 },
115 ),
116 floatingActionButton: FloatingActionButton(
117   backgroundColor: Colors.green,
118   foregroundColor: Colors.white,
119   onPressed: () {
120     Navigator.push(
121       context,
122       MaterialPageRoute(
123         builder: (_) => const TarbahPage(),
124       ),
125     );
126   },
127   child: const Text('Tambah'),
128 ),
129 );
130 }
131 }
132
133 class TarbahPage extends StatefulWidget {
134   const TarbahPage({super.key});
135
136   @override
137   State<TarbahPage> createState() => _TarbahPageState();
138 }
139
140 class _TarbahPageState extends State<TarbahPage> {
141   final _controller = TextEditingController();
142
143   @override
144   void dispose() {
145     _controller.dispose();
```

```

46
47   @override
48   Widget build(BuildContext context) {
49     return MaterialApp(
50       debugShowCheckedBanner: false,
51       title: 'Ulatir Tugas',
52       theme: ThemeData(
53         colorSchemeSeed: Colors.green,
54         useMaterial3: true,
55       ),
56       home: const TugasPage(),
57     );
58   }
59 }
60
61 class TugasPage extends StatelessWidget {
62   const TugasPage({super.key});
63
64   @override
65   Widget build(BuildContext context) {
66     final model = context.watch<TugasModel>();
67
68     return Scaffold(
69       appBar: AppBar(
70         title: Text(
71           'Tugas (${model.jumlahSelesai}/${model.items.length})',
72         ),
73         backgroundColor: Colors.green,
74         foregroundColor: Colors.white,
75       ),
76       body: model.items.isEmpty
77         ? const Center(
78           child: Text(
79             'Belum ada tugas',
80             style: TextStyle(fontSize: 18),
81           ),
82         )
83         : ListView.builder(
84           itemCount: model.items.length,
85           itemBuilder: (context, index) {
86             final tugas = model.items[index];
87
88             return ListTile(
89               leading: Checkbox(
90                 activeColor: Colors.green,
91                 value: tugas.selesai,
92                 onChanged: (_) {
93                   context.read<TugasModel>().toggle(index);
94                 },
95             ),

```

```

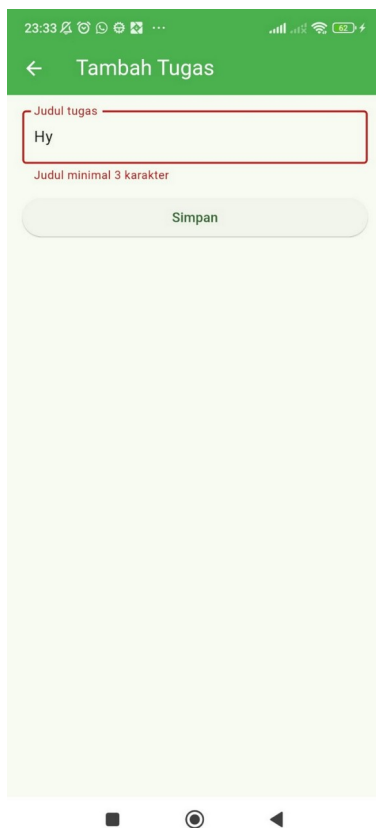
146     super.dispose();
147   }
148
149   void _simpan() {
150     final judul = controller.text.trim();
151
152     if (judul.isEmpty) {
153       return;
154     }
155
156     context.read<TugasModel>().tambah(judul);
157
158     Navigator.pop(context);
159   }
160
161   @override
162   Widget build(BuildContext context) {
163     return Scaffold(
164       appBar: AppBar(
165         title: const Text('Tambah Tugas'),
166         backgroundColor: Colors.green,
167         foregroundColor: Colors.white,
168       ),
169       body: Padding(
170         padding: const EdgeInsets.all(16),
171         child: Column(
172           children: [
173             TextField(
174               controller: _controller,
175               autofocus: true,
176               decoration: const InputDecoration(
177                 labelText: 'Judul tugas',
178                 border: OutlineInputBorder(),
179               ),
180               onSubmitted: (_) => _simpan(),
181             ),
182             const SizedBox(height: 12),
183             SizedBox(
184               width: double.infinity,
185               child: ElevatedButton(
186                 onPressed: _simpan,
187                 child: const Text('Simpan'),
188               ),
189             ),
190           ],
191         ),
192       ),
193     );
194   }
195 }
196
197 }

```

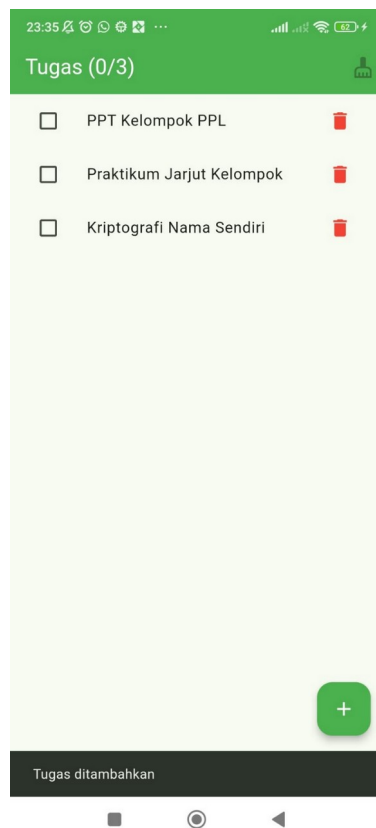
## Penjelasan

|                                             |                                                                                                                     |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| import<br>'package:provider/provider.dart'; | Mengimpor package provider supaya aplikasi bisa memakai state management berbasis Provider.                         |
| extends ChangeNotifier                      | Menjadikan TugasModel sebagai class yang dapat memberitahu widget ketika data berubah.                              |
| notifyListeners()                           | Memberitahu semua widget yang sedang mendengarkan TugasModel bahwa data berubah sehingga UI perlu diperbarui.       |
| ListView.builder                            | Membuat daftar widget secara dinamis berdasarkan jumlah data. Cocok untuk daftar tugas yang jumlahnya bisa berubah. |
| trailing                                    | Menempatkan widget di bagian kanan ListTile. Pada kode ini berisi tombol hapus.                                     |
| model.items.isEmpty ? ... : ...             | Menentukan tampilan berdasarkan kondisi. Jika daftar kosong tampil Belum ada tugas, jika ada maka tampil ListView.  |
| autofocus: true                             | Membuat TextField otomatis aktif dan langsung memunculkan keyboard ketika halaman dibuka.                           |
| Navigator.push()                            | Membuka halaman baru dan menambahkannya ke navigation stack.                                                        |

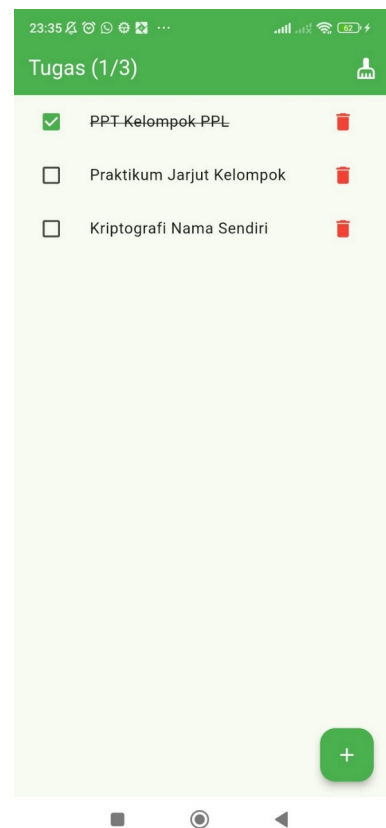
## Latihan Mandiri Modul 3



Tes yang tidak boleh harus sesuai syarat



List tugas yang di tambahkan



Tugas yang selesai atau yang di coret bisa di hapus lewat icon pembersih di pojok kanan

## Code

```

1 import 'package:flutter/material.dart';
2 import 'package:provider/provider.dart';
3
4 class Tugaz {
5   String judul;
6   bool selected;
7
8   Tugaz(this.judul, {this.selected = false});
9 }
10
11 class TugazModel extends ChangeNotifier {
12   final List<Tugaz> _tugaz = [];
13
14   List<Tugaz> get tugaz => List.unmodifiable(_tugaz);
15
16   int get jumlahSelected =>
17     _tugaz.where((tugaz) => tugaz.selected).length;
18
19   void tambah(String judul) {
20     _tugaz.add(Tugaz(judul));
21     notifyListeners();
22   }
23
24   void toggle(int index) {
25     _tugaz[index].selected = !_tugaz[index].selected;
26     notifyListeners();
27   }
28
29   void hapus(int index) {
30     _tugaz.removeAt(index);
31     notifyListeners();
32   }
33
34   void hapusSemua() {
35     _tugaz.removeWhere((tugaz) => tugaz.selected);
36
37     notifyListeners();
38   }
39 }
40
41 void main() {
42   runApp(
43     ChangeNotifierProvider(
44       create: () => TugazModel(),
45       child: const MyApp(),
46     ),
47   );
48 }
49
50 class MyApp extends StatelessWidget {
51   const MyApp({super.key});
52
53   @override
54   Widget build(BuildContext context) {
55     return MaterialApp(
56       debugShowCheckedStackBanner: false,
57       title: 'Balok Tugaz',
58       theme: ThemeData(
59         colorSchemeSeed: Colors.green,
60         useMaterial3: true,
61       ),
62       home: const TugazPage(),
63     );
64   }
65 }
66
67 class TugazPage extends StatelessWidget {
68   const TugazPage({super.key});
69
70   @override
71   Widget build(BuildContext context) {
72     final model = context.watch<TugazModel>();
73
74     return Scaffold(
75       appBar: AppBar(
76         backgroundColor: Colors.green,

```

```

36     notifyListeners();
37   },
38 );
39
40 void main() {
41   runApp(
42     ChangeNotifierProvider(
43       create: () => TugazModel(),
44       child: const MyApp(),
45     ),
46   );
47 }
48
49 class MyApp extends StatelessWidget {
50   const MyApp({super.key});
51
52   @override
53   Widget build(BuildContext context) {
54     return MaterialApp(
55       debugShowCheckedStackBanner: false,
56       title: 'Balok Tugaz',
57       theme: ThemeData(
58         colorSchemeSeed: Colors.green,
59         useMaterial3: true,
60       ),
61       home: const TugazPage(),
62     );
63   }
64 }
65
66 class TugazPage extends StatelessWidget {
67   const TugazPage({super.key});
68
69   @override
70   Widget build(BuildContext context) {
71     final model = context.watch<TugazModel>();
72
73     return Scaffold(
74       appBar: AppBar(
75         backgroundColor: Colors.green,

```

```

117       style: TextStyle(
118         decoration: Tugaz.selected
119           ? TextDecoration.lineThrough
120           : null,
121       ),
122     ),
123     trailing: Icon(Icons.
124       toggle: model.selected,
125       color: Colors.red,
126     ),
127   ),
128   onPressed: () {
129     context.read<TugazModel>().hapus(index);
130   },
131 ),
132 ),
133 ),
134 ),
135 FloatingActionButton: FloatingActionButton(
136   backgroundColor: Colors.pink,
137   foregroundColor: Colors.white,
138   onPressed: () async {
139     final result = await Navigator.pushScreen(
140       context,
141       MaterialPageRoute(
142         builder: () => const TambahPage(),
143       ),
144     );
145
146     if (result == true && context.mounted) {
147       final messenger = context.find<Messenger>();
148       const snackbar(
149         context: Text('Tugaz ditambahkan'),
150       ),
151     );
152   },
153 ),
154 ),
155 ),
156 );

```

```

157 ]
158 }
159
160 class TambahPage extends StatefulWidget {
161   const TambahPage({super.key});
162
163   @override
164   State<TambahPage> createState() => TambahPageState();
165 }
166
167 class TambahPageState extends State<TambahPage> {
168   final _formKey = GlobalKey<FormState>();
169   final _controller = TextEditingController();
170
171   @override
172   void dispose() {
173     _controller.dispose();
174     super.dispose();
175   }
176
177   void _tambah() {
178     if (_formKey.currentState!.validate()) {
179       context.read<TugazModel>().tambah(
180         _controller.text.trim(),
181       );
182
183       Navigator.pop(context, true);
184     }
185   }
186
187   @override
188   Widget build(BuildContext context) {
189     return Scaffold(
190       appBar: AppBar(
191         title: const Text('Tambah Tugaz'),
192         backgroundColor: Colors.green,
193         foregroundColor: Colors.white,
194       ),
195       body: Padding(
196         padding: const EdgeInsets.all(16),

```

```

76 foregroundColor: Colors.white,
77 title: Text(
78   'Tugas: ${model.jumlahSelesai}/${model.items.length}',
79 ),
80 actions: [
81   IconButton(
82     onPressed: 'Tugas tugas selesai',
83     icon: const Icon(Icons.checking_services),
84     onPressed: model.jumlahSelesai > 0
85       ? () {}
86       : context.read<TugasModel>().hapusSelesai(),
87     iconSize: 24,
88   ),
89 ],
90 body: model.items.isEmpty
91   ? const Center(
92     child: Text(
93       'Belum ada tugas',
94       style: TextStyle(
95         fontSize: 20,
96         fontWeight: FontWeight.bold,
97       ),
98     ),
99   )
100   : ListView.builder(
101     itemCount: model.items.length,
102     itemBuilder: (context, index) {
103       final tugas = model.items[index];
104
105       return ListTile(
106         leading: Checkbox(
107           activeColor: Colors.green,
108           value: tugas.selesai,
109           onChanged: () {} {
110             context.read<TugasModel>().toggle(index);
111           },
112         ),
113         title: Text(
114           tugas.judul,

```

```

107 child: Form(
108   key: formKey,
109   child: Column(
110     children: [
111       TextFormField(
112         controller: _controller,
113         autofocus: true,
114         decoration: const InputDecoration(
115           labelText: 'Judul tugas',
116           border: OutlineInputBorder(),
117         ),
118         validator: (value) {
119           if (value!.trim().length < 3) {
120             return 'Judul tugas wajib lebih dari 3';
121           }
122
123           if (judul.isEmpty) {
124             return 'Judul tugas wajib diisi';
125           }
126
127           if (judul.length < 30) {
128             return 'Judul maksimal 30 karakter';
129           }
130
131           return null;
132         },
133         onFieldSubmitted: () => submit(),
134       ),
135       const SizedBox(height: 12),
136       ElevatedButton(
137         width: double.infinity,
138         child: ElevatedButton(
139           onPressed: _submit,
140           child: const Text('Simpan'),
141         ),
142       ),
143     ],
144   ),
145 ),
146 ],
147 ),
148 );

```

## Penjelasan

|                                                 |                                                                                                                                |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| void hapusSelesai()                             | Method khusus untuk menghapus semua tugas yang statusnya sudah selesai.                                                        |
| _items.removeWhere((tugas) => tugas.selesai)    | Menghapus semua item dari list yang memenuhi kondisi tertentu. Pada kode ini, semua tugas dengan selesai == true akan dihapus. |
| actions: []                                     | Menambahkan widget/tombol pada bagian kanan AppBar.                                                                            |
| onPressed: model.jumlahSelesai > 0 ? ... : null | Tombol hanya aktif jika ada minimal satu tugas selesai. Jika tidak ada, onPressed menjadi null dan tombol nonaktif.            |
| onPressed: () async { ... }                     | Menandakan callback bersifat asynchronous sehingga bisa menggunakan await.                                                     |
| final berhasil = await ...                      | Menyimpan hasil yang dikembalikan dari halaman TambahPage. Pada kode ini hasilnya true jika tugas berhasil ditambahkan.        |
| onFieldSubmitted                                | Event pada TextFormField yang berjalan ketika pengguna menekan tombol submit/enter pada keyboard.                              |
| berhasil == true && context.mounted             | Memastikan tugas berhasil ditambahkan dan halaman masih aktif sebelum menampilkan Snackbar.                                    |

## Tugas Daftar Belanja

Tambah Barang

Nama Barang

Nama barang wajib diisi

#Jumlah

Jumlah wajib diisi

Kategori

Kategori wajib dipilih

Simpan Barang

Harus input sesuai syarat

Daftar Belanja (5 belum dibeli)

☐

Sabun

Jumlah: 10 • Kebutuhan Rumah

☐

Beras

Jumlah: 3 • Kebutuhan Rumah

☐

Telur

Jumlah: 100 • Makanan

☐

Sayur

Jumlah: 2 • Makanan

☐

Aqua

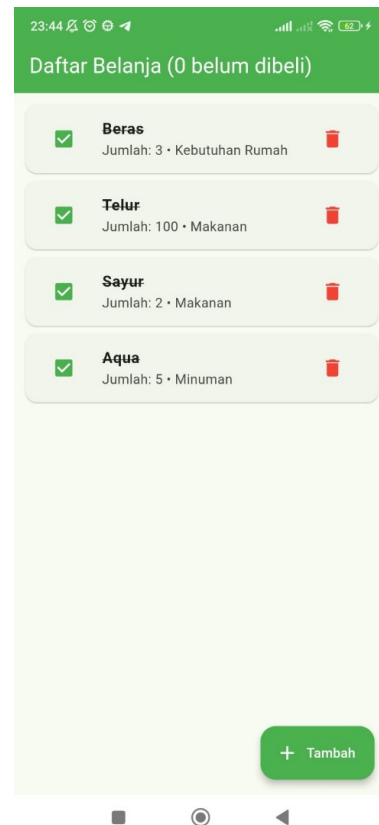
Jumlah: 5 • Minuman

+ Tambah

List Daftar belanja yang belum di beli



List daftar belanja yang sudah di beli di coret



Daftar belanja yang di hapus

## Code

```

1  from . import plot, visualization
2  from . import plot_helpers, visualization_helpers
3
4  class Curve:
5      """Base class for all curves"""
6      def __init__(self):
7          self._x = None
8          self._y = None
9          self._x_labels = None
10         self._y_labels = None
11
12     def __str__(self):
13         """String representation of the curve"""
14         return self.__class__.__name__
15
16     def __repr__(self):
17         """String representation of the curve"""
18         return self.__class__.__name__
19
20     def __eq__(self, other):
21         """Equality comparison"""
22         return self.__class__.__name__ == other.__class__.__name__
23
24     def __neq__(self, other):
25         """Inequality comparison"""
26         return self.__class__.__name__ != other.__class__.__name__
27
28     def __lt__(self, other):
29         """Less than comparison"""
30         return self.__class__.__name__ < other.__class__.__name__
31
32     def __gt__(self, other):
33         """Greater than comparison"""
34         return self.__class__.__name__ > other.__class__.__name__
35
36     def __le__(self, other):
37         """Less than or equal to comparison"""
38         return self.__class__.__name__ <= other.__class__.__name__
39
40     def __ge__(self, other):
41         """Greater than or equal to comparison"""
42         return self.__class__.__name__ >= other.__class__.__name__
43
44     def __add__(self, other):
45         """Addition"""
46         return self.__class__.__name__ + other.__class__.__name__
47
48     def __sub__(self, other):
49         """Subtraction"""
50         return self.__class__.__name__ - other.__class__.__name__
51
52     def __mul__(self, other):
53         """Multiplication"""
54         return self.__class__.__name__ * other.__class__.__name__
55
56     def __div__(self, other):
57         """Division"""
58         return self.__class__.__name__ / other.__class__.__name__
59
60     def __mod__(self, other):
61         """Modulo"""
62         return self.__class__.__name__ % other.__class__.__name__
63
64     def __pow__(self, other):
65         """Power"""
66         return self.__class__.__name__ ** other.__class__.__name__
67
68     def __call__(self, x):
69         """Call method"""
70         return self.__class__.__name__(x)
71
72     def __getitem__(self, x):
73         """Indexing"""
74         return self.__class__.__name__[x]
75
76     def __setitem__(self, x, y):
77         """Indexing"""
78         self.__class__.__name__[x] = y
79
80     def __delitem__(self, x):
81         """Indexing"""
82         del self.__class__.__name__[x]
83
84     def __len__(self):
85         """Length"""
86         return len(self.__class__.__name__)
87
88     def __iter__(self):
89         """Iteration"""
90         return iter(self.__class__.__name__)
91
92     def __reversed__(self):
93         """Reverse iteration"""
94         return reversed(self.__class__.__name__)
95
96     def __contains__(self, x):
97         """Contains"""
98         return x in self.__class__.__name__
99
100    def __contains__(self, x):
101        """Contains"""
102        return x in self.__class__.__name__

```

```

165     for (right = 0; !is; !isRightFound) {
166         isLeft = false;
167         for (left = 0; !is; !isLeftFound) {
168             isLeft = true;
169             break;
170         }
171         if (isLeft) {
172             isLeft = false;
173             break;
174         }
175         if (isLeft) {
176             isLeft = false;
177             break;
178         }
179         if (isLeft) {
180             isLeft = false;
181             break;
182         }
183     }
184     if (isLeft) {
185         isLeft = false;
186         break;
187     }
188     if (isLeft) {
189         isLeft = false;
190         break;
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204     if (isLeft) {
205         isLeft = false;
206         break;
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209         isLeft = false;
210         break;
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630         break;
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634         break;
635     }
636     if (isLeft) {
637         isLeft = false;
638         break;
639     }
640     if (isLeft) {
641         isLeft = false;
642         break;
643     }

```

[illegible][illegible]

[illegible]

|                                       |                                                                                                  |
|---------------------------------------|--------------------------------------------------------------------------------------------------|
| FloatingActionButton.extended         | Versi FloatingActionButton yang bisa memiliki <b>ikon dan teks sekaligus</b> .                   |
| hintText                              | Memberikan contoh atau petunjuk input di dalam field sebelum pengguna mengetik.                  |
| prefixIcon                            | Menambahkan icon di bagian kiri input field.                                                     |
| keyboardType:<br>TextInputType.number | Menampilkan keyboard angka pada perangkat mobile agar pengguna lebih mudah memasukkan jumlah.    |
| int.tryParse(value)                   | Mencoba mengubah input teks menjadi integer. Jika gagal, hasilnya null dan aplikasi tidak crash. |
| padding: const EdgeInsets.all(8)      | EdgeInsets.all(8) Memberikan jarak di sekeliling isi ListView.                                   |
| Icons.shopping_cart_outlined          | Icon keranjang belanja berbentuk outline untuk tampilan saat daftar masih kosong.                |